

CHE 565 -- Homework 3

Soki Sem

April 18, 2026

Problem 1

Steepest Descent

Function :

$$f(x_1, x_2) = 0.5x_1^2 + 1.0x_1x_2 - x_1 + 1.0x_2^2 - x_2$$

At $x_0 = \text{Matrix}([1], [1])$, $f(x_0) = 0.5000000000000000$

Gradient of f:

$$\nabla f = \begin{bmatrix} 1.0x_1 + 1.0x_2 - 1.0 \\ 1.0x_1 + 2.0x_2 - 1 \end{bmatrix}$$

At $x_0 = \text{Matrix}([1], [1])$, $Df(x_0) = 1.00, 2.00$

$$\begin{bmatrix} g_1(\alpha) \\ g_2(\alpha) \end{bmatrix} = \begin{bmatrix} 1 - 1.0\alpha \\ 1 - 2.0\alpha \end{bmatrix}$$

Function phi_k:

$$\phi(\alpha) = 6.5\alpha^2 - 5.0\alpha + 0.5$$

Derivative of phi_k:

$$\begin{aligned} \frac{d}{d\alpha}\phi(\alpha) &= 13.0\alpha - 5.0 \\ 0 &= 13.0\alpha - 5.0 \end{aligned}$$

Optimal alpha: 0.38

Next iterate x1: 0.62, x2: 0.23

Function value at next iterate: -0.46

Problem 2

Conjugate Gradient Method

$$\begin{bmatrix} s_1(\alpha) \\ s_2(\alpha) \end{bmatrix} = \begin{bmatrix} -1.0x_3 - 1.0x_4 + 2(-1.0x_3 - 2.0x_4 + 1.0)(1.0x_3 + 2.0x_4 - 1) + 2(-1.0x_3 - 1.0x_4 + 1.0)(1.0x_3 + 1.0x_4 - 1) + 1 \\ -1.0x_3 - 2.0x_4 + 2(-2.0x_3 - 4.0x_4 + 2.0)(1.0x_3 + 2.0x_4 - 1) + 2(-2.0x_3 - 2.0x_4 + 2.0)(1.0x_3 + 1.0x_4 - 1) + 1 \end{bmatrix}$$